

Worksheet

Do the following series converge or diverge? Be sure to state any tests used and show all work needed apply them.

1.
$$\sum_{k=0}^{\infty} \frac{4k}{5k-3}$$

2.
$$\sum_{k=1}^{\infty} \frac{\ln(k^2)}{k}$$

3.
$$\sum_{k=1}^{\infty} \sqrt[k]{k}$$

4.
$$\sum_{k=1}^{\infty} \frac{3}{k\sqrt[3]{k}}$$